

Assignment 1

1. Mean, Median, Mode

- Given a dataset of student marks:
45, 50, 55, 60, 60, 65, 70
 - Calculate mean, median, and mode.
 - Interpret which measure best represents the dataset.

2. Range

- Find the range of the dataset:
12, 18, 25, 30, 45, 60
- Explain what the range indicates about data spread.

3. Quantiles

- Divide the dataset into quartiles:
10, 20, 30, 40, 50, 60, 70, 80
- Identify Q1, Q2, Q3.

4. Interquartile Range (IQR)

- Using the above dataset:
 - Calculate IQR
 - Identify if any outliers exist

5. Percentiles

- Calculate the 25th and 90th percentile of:
5, 10, 15, 20, 25, 30, 35, 40

6. Variance

- Calculate variance for:
4, 8, 6, 5, 3
- Explain what variance tells about the data.

7. Types of Variance

- Compute:
 - Population variance
 - Sample variance
- Use dataset: 2, 4, 6, 8

8. Standard Deviation

- Find standard deviation for:
10, 20, 30, 40, 50
- Interpret the result.

9. Coefficient of Variation

- Given:
 - Mean = 50
 - Standard deviation = 5
- Calculate coefficient of variation and explain its significance.

10. Univariate vs Bivariate Analysis

- Given:
 - Dataset 1: Student marks
 - Dataset 2: Marks vs Study hours
- Identify which is univariate and which is bivariate
- Explain with reasoning

11. Covariance and Correlation

- Given two variables:
 - $X = [1, 2, 3, 4]$
 - $Y = [2, 4, 6, 8]$
- Calculate covariance and correlation
- Interpret the relationship

12. Resampling

- Explain bootstrap resampling
- Perform a simple resampling (random sampling with replacement) on:
10, 20, 30, 40

13. Sample Space and Events

- Toss a coin twice
- Write:
 - Sample space
 - Possible events

14. Types of Events

- Identify:
 - Mutually exclusive events
 - Independent events
- Use dice example

15. Joint Probability

- Probability of:
 - Getting a Head and rolling a 6
- Show calculation

16. Conditional Probability

- Given:
 - 30 students, 10 like math, 5 like both math and science
- Find probability that a student likes science given they like math

17. Marginal Probability

- From a given probability table:
 - Calculate marginal probabilities

18. Bayes' Theorem

- A disease test:
 - 1% population has disease
 - Test accuracy = 95%
- Calculate probability a person has disease given a positive test

19. Data Analytics Lifecycle

- Explain stages:
 - Data collection
 - Cleaning
 - Analysis
 - Visualization
- Apply on a real-world case:
 - Sales data analysis

20. Visualization using Power BI

- Create a dashboard in Power BI:
 - Import sample sales dataset
 - Create:
 - Bar chart
 - Pie chart
 - Line chart
- Write insights from dashboard